

PRICE AND MARKET SEGMENTATION CONSEQUENCES OF TARGETED VOUCHERS*

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Abstract

We study how a large targeted voucher reform reshapes tuition-setting incentives and market segmentation in competitive education markets. We focus on Chile’s targeted voucher program, introduced in 2008, which increased per-student funding for disadvantaged pupils while prohibiting participating schools from charging top-up fees to eligible students; participation was voluntary. Using administrative data covering the universe of schools and students over 2004–2015, we exploit cross-municipality variation in pre-reform eligibility shares in an event-study design to estimate equilibrium effects on fees and program take-up. We find that the reform induces a segmented price schedule: eligible students face a mechanical drop to zero fees in participating schools, while fees faced by non-eligible students fall on average in more exposed markets, consistent with participating schools cutting posted top-up fees for higher-income families. At the same time, private-voucher schools that opt out raise fees, moving upmarket. We interpret these patterns through models of school competition with peer effects and sorting, and we provide supporting evidence from a pre-reform school choice model with peer composition that families exhibit economically meaningful willingness-to-pay to avoid low-income peers.

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1 Introduction

Targeted subsidies are a central tool in education policy. By directing additional public resources toward disadvantaged students, governments aim to raise achievement and narrow socioeconomic gaps while preserving the advantages of parental choice and school autonomy. Yet targeted vouchers can also reshape competition among schools because they change schools’ effective marginal revenue and the composition of demand they face. When schools can charge top-up fees, a targeted voucher may alter incentives to set prices differently across income groups and to sort students through both prices and program participation decisions. Understanding these equilibrium responses is essential for evaluating targeted voucher reforms: the distribution of prices that families face and the degree of market segmentation they induce are themselves core outcomes, and they are key mechanisms through which the policy affects access, sorting, quality, and welfare.

This paper studies the effects of Chile’s *Ley de Subvención Escolar Preferencial (SEP)*—a large-scale targeted voucher introduced in 2008—on schools’ pricing decisions and on market segmentation in primary education. Chile is a natural setting for this question. Since the early 1980s, the country has operated a nationwide voucher system in which public schools and private-voucher schools receive a per-student subsidy, and private-voucher schools may additionally charge a top-up fee. The SEP reform increased per-student funding for economically disadvantaged pupils and introduced a key constraint for participating schools: they could not charge top-up fees to eligible (low-income) students. Participation was voluntary for public and private-voucher schools. As a result, after 2008 the market could feature two price schedules at the same school: a mandated zero fee for eligible students in participating schools and a positive posted fee for non-eligible students.

We ask three related questions. First, how did the targeted voucher reform affect the prices (top-up fees) that schools charged, and did these effects differ across income groups? Second, did the reform lead to market segmentation, in the sense that traditionally low-quality, low-fee schools opted into the program and further reduced their fees, while high-quality, high-fee schools opted out and increased prices? Third, can standard mechanisms in models of school competition—in particular, the interaction between prices and peer composition—help explain the observed pricing patterns?

Our empirical strategy exploits two key features of the policy. The reform was implemented nationwide and unexpectedly, creating a sharp time shock in 2008. At the same time, the reform’s potential impact differed across local education markets because municipalities varied substantially in the share of students eligible for the targeted voucher. Following Hsieh and Urquiola (2006) and Sánchez (2019), we define markets at the municipality level and measure each market’s *intensity of treatment* as the share of eligible students in the municipality one year before the reform. We then estimate an event-study design that interacts calendar time relative to 2008 with this pre-reform intensity measure. This design captures equilibrium effects operating through both sides of the market: changes in pricing, changes in participation decisions, and changes in student sorting.

We combine administrative data from the Chilean Ministry of Education covering the universe of schools offering primary grades with multiple complementary datasets on students, schools, and teachers for 2004–2015. We observe school-level top-up fees, enrollment, participation in the targeted voucher program, and a rich set of school inputs. These data allow us to distinguish between the fee schedule faced by low-income (eligible) students and that faced by higher-income (non-eligible) students, which is central for interpreting pricing responses under the reform.

Our headline result is that the targeted voucher reform induced a strong segmentation of the fee distribution faced by students. For low-income (eligible) students, the policy mechanically pushed prices down to zero in participating schools, expanding access to previously fee-charging private-voucher schools. For higher-income students, we find that the average fee they face falls in more exposed markets, consistent with participating schools reducing the top-up fees charged to non-eligible students. At the same time, private-voucher schools that opted out of the program increased their posted top-up fees, moving upmarket. Taken together, these responses imply that the reform did not simply shift the level of fees; it changed the *structure* of prices in a way that is consistent with equilibrium market segmentation.

Our reduced-form findings map naturally into the theoretical predictions of models of school competition with peer effects (Rothschild and White, 1995; Eppele and Romano, 1998; Eppele et al., 2006). In these frameworks, households value peer composition and schools use tuition and other instruments to sort students, so equilibrium responses to a policy that differentially changes the effective price of enrolling disadvantaged students can take the form of increased differentiation across schools and segmented price schedules. Consistent with this logic, we find that the targeted voucher induces opposing fee movements across income groups and changes the structure of posted fees through voluntary program participation. Our preference estimates from a pre-reform school choice model with peer composition (Appendix C) provide direct support for the key mechanism: families exhibit economically meaningful willingness-to-pay to avoid low-income peers, implying that schools have incentives to adjust prices for non-eligible families to offset peer-composition disutility when participation increases the share of disadvantaged students.

This paper contributes to the literature on vouchers, school competition, and the equilibrium effects of targeted education policies by documenting how a redistributive reform can reshape the *price schedule* faced by different socioeconomic groups. First, we complement work on competition and sorting under generalized school choice systems (Hsieh and Urquiola, 2006) by focusing on a targeted voucher implemented in a context where schools can charge top-up fees and can choose whether to participate. In this environment, the relevant object for distributional analysis is not a single posted tuition, but a menu of prices that depends on program participation and student eligibility. We show that targeted vouchers generate a sharp, policy-induced wedge in feasible pricing—zero fees for eligible students in participating schools alongside positive fees for non-eligible students—and that schools respond endogenously by adjusting the fees charged to non-eligible students and by sorting into (in/out) regimes. These equilibrium price responses yield segmentation patterns that are distinct from, and potentially interact with, the longer-run achievement and inequality impacts of Chile’s targeted voucher program studied in Correa et al. (2014), Navarro-Palau (2017), Sánchez, 2019, Aguirre (2020), Sánchez (2022), Gazmuri (2024), and Neilson (2025), among others.

Second, our results speak to the broader voucher evidence base that has largely emphasized individual-level outcomes and partial-equilibrium effects. In influential randomized voucher settings such as Colombia (Angrist et al., 2002, 2006), vouchers can raise attainment and test scores; our contribution is to show how, in a nationwide program with voluntary school participation and top-up fees, a targeted subsidy can also induce strategic supply responses that reallocate who faces which prices. By estimating market-level event-study effects that incorporate both demand-side sorting and supply-side participation and pricing, we provide direct evidence on the equilibrium mechanisms that mediate access and stratification when vouchers operate through competitive school markets.

Third, we connect reduced-form evidence on fees to mechanisms emphasized in canonical models of school competition with peer effects (Rothschild and White, 1995; Eppele and Romano, 1998; Eppele et al., 2006).

We interpret the observed fee changes as consistent with strategic complementarities between prices and peer composition: after the reform, schools serving a higher share of disadvantaged students may lower fees for non-eligible families to offset peer-related disutility, while schools opting out can raise fees as their demand becomes relatively less price elastic. Our pre-reform school choice estimates provide supporting evidence for this channel by revealing economically meaningful willingness-to-pay to avoid low-income peers.

The remainder of the paper proceeds as follows. Section 2 describes Chile’s school system and the targeted voucher program. Section 3 presents the administrative data. Section 4 lays out the identification strategy and reports the empirical results on fees and related outcomes. Section 5 concludes.

2 The Chilean School System and the Targeted Voucher Program

Schools in Chile can be organized within three main groups according to their management and financing scheme: public schools, private-voucher schools, and private-non-voucher schools. Both public and private-voucher schools are financed by per-student voucher subsidies paid by the government directly to the schools. In addition to the subsidies, private-voucher schools are allowed to charge fees on top of the vouchers they receive. Private-non-voucher schools are entirely financed by fees charged to parents, and serve the country’s richest families. Today, 38% of students in primary education grades attend public schools, 53% attend private-voucher schools, and 9% attend private-non-voucher schools.

Since 1981, every student in Chile is entitled to an individual voucher to be used to (at least partially) cover tuition at any public or private-voucher school of her choice. In 2008, the government introduced a new source of subsidy to complement the existing flat voucher in the form of a targeted voucher to economically disadvantaged students. In February of that year, the *Ley de Subvención Escolar Preferencial (SEP)* law that regulates this new subsidy was enacted and immediately put into practice for the 2008 academic year.¹ The law mandates that each school that participates in the program receives an additional subsidy per every eligible low income student that it enrolls, with the requirement of not charging top-up fees to those students. In addition, each participating school also receives a per-student subsidy that depends on the share of low income students enrolled in the school, called *Subvención por Concentración*, or concentration subsidy. Participation in the program is voluntary on the part of schools, and only public and private-voucher schools are eligible to join. Participating schools are required to set short- and long-term learning goals (i.e. test score achievements), which are evaluated by the government every four years.

Table 1 displays the evolution of the monthly per-student voucher subsidies corresponding to elementary grades 1st–4th, decomposed by its different categories, for the years 2004–2015. The voucher amounts are in US\$ as of May 2016, and correspond to schools with full day shifts. The universal voucher, which is set to represent the unit cost of educating a student, starts at \$63 in 2004 and steadily increases to reach \$93 in 2015. The targeted voucher is set to represent the additional unit cost of educating a low income student, and starts at \$37 in 2008, reaching up to \$57 in 2015. The concentration subsidy is considerably lower, and stays fairly constant over time at about 4% and 10% the universal voucher, depending on the share of eligible low income students enrolled in the school.

¹The yearly academic calendar in Chile starts in March and ends in December.

Table 1: Monthly Voucher Subsidy Decomposition for Students in 1st–4th Grades

category	subsidy (US\$)											
	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015
universal voucher	62.66	63.07	64.92	63.82	62.40	80.86	82.48	82.71	84.77	91.50	90.93	92.62
targeted voucher	–	–	–	–	36.81	42.86	43.71	43.84	44.93	55.93	55.59	56.62
concentration subsidy:												
15–30%	–	–	–	–	2.66	3.00	3.06	3.06	3.14	3.89	3.87	3.94
30–45%	–	–	–	–	4.56	5.14	5.24	5.26	5.39	6.67	6.62	6.75
45–60%	–	–	–	–	6.08	6.85	6.99	7.01	7.18	8.88	8.82	8.99
more than 60%	–	–	–	–	6.84	7.71	7.86	7.89	8.08	9.97	9.91	10.09

Notes: All values are real, and are converted from Chilean pesos to US dollars according to the exchange rate of Ch\$686.52 per US dollar, as of May 16, 2016. The universal voucher values correspond to students at schools with full day shifts.

The criteria to classify a student as being eligible to be a beneficiary of the targeted voucher program was set to cover households in the lowest 40% of the income distribution. The eligibility criteria include various components, but the two most important are: (1) the household belongs to the lowest 33% of a socioeconomic index distribution used by the government to guide the delivery of social programs (i.e. *Ficha de Protección Social*), and (2) the household participates in *Chile Solidario*, a social program that serves families in vulnerable conditions.² By 2011, about 48% of students in the country were classified to be eligible to receive the targeted voucher, and 88% of them (42% of all students) enrolled in participating schools (either a public school or a private-voucher school that joined the program). In other words, the program impacts almost half of the student population in the country.

In line with its motivating goals, the targeted voucher reform was able to expand access to ex-ante more expensive private schools for some low-income students. When compared to the year preceding the introduction of the program, where more than 75% of low-income students attended zero-fee schools, in 2013, the 75th percentile of vulnerable students enrolled in a school that charged \$10.3 per month in the pre-reform period. Moreover, while 58% and 42% of low-income students attended public and subsidized private schools in 2007, respectively, this enrollment distribution shifted towards the private sector six years after the start of the reform, with 48% and 52% of low-income students enrolling in public and subsidized private schools, respectively.

The reform also induced a sharp divergence in pricing behavior among private-voucher schools depending on whether they opted into the program. Before the reform, schools effectively charged a single posted fee; among eventual participants, this fee averaged \$12 per month. Six years after the introduction of the reform, participating schools charged zero top-up fees to low-income students and charged \$9.9 to high-income students. In contrast, schools that opted out of the program increased posted fees from \$61.6 before the reform to \$74.6 after the reform. Together, these patterns are suggestive of a market segmentation response, with participating schools becoming relatively more affordable to low-income students while non-participating schools moving upmarket.

²See Aguirre (2020) and Neilson (2025) for more details on the criteria to classify a student to be eligible to receive the targeted voucher.

3 Data

We combine various administrative data sets of Chilean students and schools for the years 2004–2015 to form a twelve-year (unbalanced) panel sample for schools, and a twelve-year (unbalanced) panel sample for municipalities. The data were obtained from the Ministry of Education, and include the censuses of students, schools, and teachers.³ Our analysis sample consists of the universe of Chilean schools that offer any of 1st–8th grades (i.e. primary education).

Table 2 displays the total number of schools offering primary level grades, by type and participation in the targeted voucher program status, for the years 2004–2015. Panel A shows the number of schools, overall and by type. The total number of schools decreases slightly over time, going from 8,908 schools in 2004 to 8,425 in 2015, possibly reflecting the recent changes in demographics and family size preferences in Chile. Public and private-voucher schools represent about 94% of all schools in the country, highlighting the important (ex-ante) potential reach of the targeted voucher program. Public schools are the most numerous, but they importantly decrease their presence over the period studied, going from 5,426 in 2004 to 4,613 in 2015. Private-voucher schools, in contrast, see an increase in their presence in the system, going from 2,928 in 2004 to 3,386 in 2015. This observed pattern for public and private-voucher schools is consistent with estimated models of families’ preferences for school types in Chile.⁴ Private non-voucher schools represent a small fraction of all schools, and are 554 in 2004, and 426 in 2015. Panel B displays the number of schools that participate in the targeted voucher program, overall and by type. Almost all public schools (97%) immediately join the program in 2008, the year of its introduction. Public schools’ participation in the program remains almost universal throughout the period, reaching 99% in 2015. Private-voucher schools’ participation is also important in 2008, with 47% of these schools joining the program. Private-voucher schools’ participation rapidly increases over time, reaching 75% in 2015. These figures highlight the important ex-post reach of the targeted voucher program in the Chilean system of education.

Table 2: Schools by Type and Participation in the Targeted Voucher Program

	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015
<i>A. All</i>	8,908	8,924	8,871	8,839	8,827	8,848	8,804	8,699	8,674	8,573	8,491	8,425
public	5,426	5,397	5,267	5,192	5,126	5,080	5,004	4,882	4,826	4,740	4,660	4,613
private-voucher	2,928	3,068	3,159	3,202	3,259	3,336	3,365	3,390	3,425	3,414	3,409	3,386
private-non-voucher	554	459	445	445	442	432	435	427	423	419	422	426
<i>B. In Program</i>	–	–	–	–	6,503	6,864	6,912	7,011	7,150	7,133	7,111	7,125
public	–	–	–	–	4,957	4,969	4,898	4,782	4,759	4,665	4,618	4,582
private-voucher	–	–	–	–	1,546	1,895	2,014	2,229	2,391	2,468	2,493	2,543

Notes: Panel A shows the number of schools that offer primary education. Panel B shows the number of schools that offer primary education and that participate in the targeted voucher program.

We describe our analysis sample and outcomes of interest in Table 3. We keep in the final sample all school-year observations that do not present missing observations in all variables of analysis. Table 3 shows the average and the standard deviation (in parentheses) of top-up fees charged by private-voucher schools,

³See Appendix A for a more detailed description of each of the data sets we use.

⁴See Sánchez (2022), and Neilson (2025), among others.

for the years 2004–2015. Fees are in real US\$, as of May 2016. Notice that, before the introduction of the targeted voucher program, schools were allowed to charge a unique (possibly positive) fee on top of the subsidies to all students, regardless of the socioeconomic status of the student. With the start of the targeted policy, schools that decide to join the program are mandated to charge zero fees to eligible low-income students, and continue to be allowed to charge any (possibly positive) top-up fee to higher income students. Non-participating schools, in contrast, continue to be allowed to charge a unique top-up fee to all students. Thus, with the targeted reform in place, there exist two vectors of top-up fees in the market, one faced by low-income students that are eligible to be beneficiaries of the targeted voucher program, and another faced by non-eligible higher-income students. Before the reform, the average top-up fee is \$16.14 in 2004, and it slightly increases to \$17.24 in 2007. After the introduction of the targeted voucher reform, top-up fees faced by low income students are considerably reduced to about \$14–15, on average. In contrast, top-up fees charged to higher income students continue to increase slowly, reaching \$20.04 in 2015.

Table 3: Summary Statistics - Private-voucher School Top-up Fees

	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015
low income	16.14 (25.29)	17.36 (26.81)	18.34 (31.25)	17.24 (26.92)	14.46 (27.31)	14.97 (34.07)	15.04 (29.86)	14.40 (30.05)	14.96 (35.24)	15.41 (37.28)	14.95 (33.32)	14.71 (34.31)
higher income	16.14 (25.29)	17.36 (26.81)	18.34 (31.25)	17.24 (26.92)	17.82 (27.47)	19.33 (34.36)	19.29 (30.39)	18.98 (30.42)	20.49 (36.67)	20.87 (37.36)	20.13 (33.51)	20.04 (34.67)
observations	2,853	3,013	3,106	3,143	3,201	3,247	3,283	3,341	3,369	3,397	3,395	3,374

Notes: School level mean values. Standard deviations are in parentheses. All values are real, and are converted from Chilean pesos to US dollars according to the exchange rate of Ch\$686.52 per US dollar, as of May 16, 2016.

4 The Effects of Targeted Vouchers on School Prices

4.1 Identification Strategy

We follow Hsieh and Urquiola (2006) and Sánchez (2019), and use the geopolitical boundaries of municipalities to define educational markets in Chile.⁵ Enrollment data is consistent with this definition, as about 90% of students attend a school that is located in the same municipality of their residence.

The particular design of the targeted voucher program implies that some markets are more affected by the program than others, depending on their share of students that are eligible to be beneficiaries of the new subsidy (i.e. low income students). To make this point clear, take the extreme case of a market in which no eligible low income student resides. The targeted voucher reform has zero incidence in this market, in terms of adding new funds, because no student is eligible to receive the targeted subsidy. Conversely, a market in which all students come from economically disadvantaged families has the maximum potential of receiving additional funds.⁶ Thus, it is possible to argue that different markets have different intensities of treatment, and that these treatment intensities depend on the markets' share of eligible students.

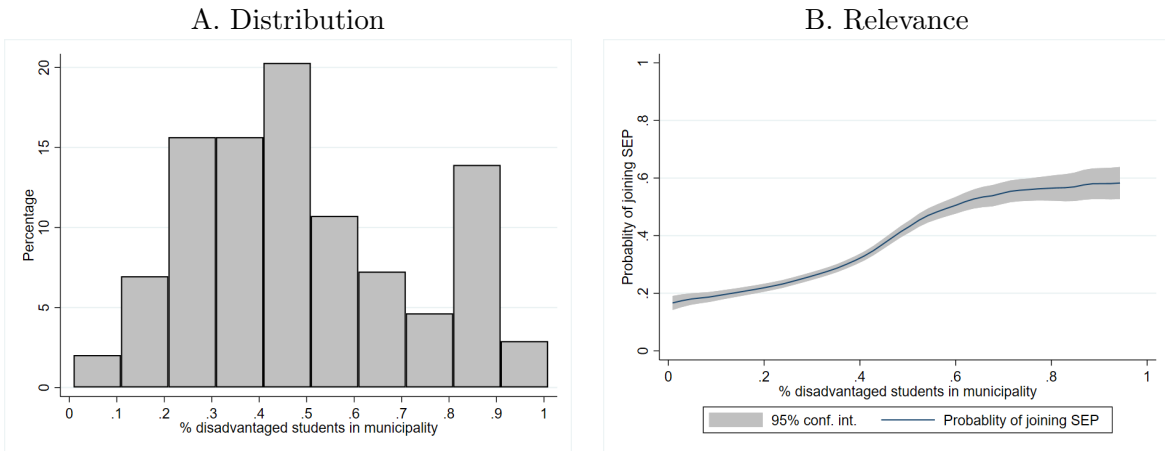
⁵See Topel (1986) and Card (2001) for other studies that use political and administrative boundaries to define local markets.

⁶We are careful to say that this increase in the funding is only potential (ex-ante) because it primarily depends on schools deciding to participate in the program. However, as we show below, the share of eligible students in the market highly predicts the likelihood that a school chooses to participate in the targeted voucher program.

To avoid endogeneity issues when conducting our empirical analysis, we use students' municipality of residence one year prior to the introduction of the targeted voucher program.⁷ This variable is highly correlated with the current municipality of residence, and is free of endogeneity because the residential decision was made before the program was announced and implemented.⁸

Figure 1 displays the distribution and relevance of the intensity-of-treatment variable. Panel A shows the distribution of municipalities according to their share of eligible students one year before the targeted voucher reform was implemented. The support of the distribution covers the full $[0, 1]$ range. About half of municipalities have between 20% and 50% of eligible low-income students, and only a few have less than 10% or more than 90% of eligible students. Panel B displays a nonparametric estimate of the probability that a private-voucher school joins the targeted voucher program in the first year of implementation (2008) as a function of the municipality's share of low-income students in 2007. The estimated function is monotonically increasing over $[0, 1]$, with the estimated probability rising from about 0.2 to about 0.6. We therefore conclude that (1) the distribution of the intensity-of-treatment variable provides enough variation for our analysis, and (2) the share of low-income students in the municipality (one year before the reform) is a strong predictor of private-voucher schools' participation decisions.

Figure 1: Distribution and Relevance of Intensity of Treatment



Notes: Panel A: distribution of markets according to their share of low income students one year prior to the introduction of the targeted voucher reform (2007). Panel B: nonparametric estimation of the probability that a private-voucher school joins the targeted voucher program in the first year of its implementation (2008), with respect to the share of low income students in the market one year prior to the introduction of the targeted voucher program (2007).

4.2 Empirical Strategy

We exploit the exogenous variation given by the (unanticipated) implementation of the targeted voucher reform, as well as the one given by the intensity of treatment in the form of the share of eligible students

⁷As opposed to the municipality of residence prior to the introduction of the program, the current municipality of residence is subject to endogeneity issues via, for example, endogenous migration (Rosenzweig and Wolpin, 1988).

⁸88% of 4th graders in 2008-2011 live in the same municipality as they did in 2007.

in the municipality (one year prior to the reform) to implement an event study design. The implementation of the event study allows us to test whether the reform has an effect on the various variables of interest presented in Table 3.

To test the effects of the reform on schools' top-up fees, we estimate the following equation,

$$y_{ist} = \gamma_s + \lambda_t + \sum_{\tau=-4}^7 \delta_{\tau}(D_{\tau t} \times \text{intensity}_s) + \varepsilon_{ist}, \quad (1)$$

where y_{ist} is the outcome of interest (fees) for school i in municipality s in period t , γ_s is a municipality fixed effect, λ_t is a year fixed effect, $D_{\tau t}$ is a year dummy variable, with $\tau = 0$ indicating the year of the introduction of the reform, intensity_s is the intensity of treatment (i.e. share of eligible low income students in the municipality of residence one year before the reform), and ε_{ist} is an error term. In the empirical implementation of this regression, we cluster the standard errors at the municipality level (Bertrand et al., 2004; Cameron and Miller, 2015). The coefficients of interest are δ_{τ} , as they capture the period effects of the targeted voucher reform.

The treatment effect parameters identified by our event study design are the average treatment effects on the treated, where the treatment corresponds to being located in a market where all students are eligible to receive the targeted voucher as opposed to being located in a market where none of the students is eligible to be a targeted voucher beneficiary. Consequently, our estimates incorporate equilibrium effects on the demand (e.g. student sorting among schools) and on the supply (e.g. schools' decision to participate in the program) sides of the market. Moreover, our results include the responses of both participating and non-participating schools.⁹

Our treatment, and therefore its interpretation, is in contrast to that of Correa et al. (2014) and Feigenberg et al. (2019), which corresponds to the effect of a school choosing to participate in the program, and does not necessarily account for equilibrium effects in the market.

4.3 Results

Before we examine the effects of the targeted voucher reform on private-voucher schools' top-up fee decisions, it is important to note that before the introduction of the targeted voucher program, all private-voucher schools were allowed to charge a unique fee to students on top of the per student subsidy they received. The reform, once implemented, did not bring any change to the pricing rules for non-participating schools. However, private-voucher schools that decided to be part of the targeted voucher program were imposed a restriction on the fees they charge to low income students, which resulted in these schools (potentially) charging differently to low and higher income students. Specifically, participating schools were mandated to charge zero top-up fees to low income students, and they continued to be allowed to charge any (possibly positive) fee to higher income students. Consequently, as long as at least one private-voucher school joins the program, two different vectors of top-up fees exist in the market: one that is faced by low income students, who are charged zero in participating schools and the sticker top-up fee in non-participating schools; and

⁹In a game-theoretic and oligopolistic competition setting, where a school's strategies impose externalities on its competitors, the targeted voucher reform impacts the decisions of schools that decide to participate in the program, as well as the strategies of schools that choose not to participate in the program. See, for instance, the model developed in Sánchez (2022).

another that is faced by higher income students, who pay the sticker top-up fee at each school, regardless of the program participation status of the school.

Thus, examining the effects of the reform on private-voucher schools' top-up fee decisions necessarily implies distinguishing between the effects on the top-up fees charged to low-income students and the effects on the top-up fees charged to higher income students.

To make things clear, let p_j be the top-up fee private-voucher school j charges in the absence of the targeted voucher reform. Consequently, $E[p_j]$ is the expected top-up fee schools charge in the absence of the reform. With the introduction of the targeted voucher program, schools must decide whether they participate in the program or not. Denote by $\tau_j \in \{0, 1\}$ school j 's decision to participate in the targeted voucher program once implemented (1 indicates participation, 0 non-participation). Also, let p_j^1 denote the counterfactual top-up fee school j charges to high income students if the school participates in the program.¹⁰ Similarly, let p_j^0 denote the counterfactual top-up fee school j charges to all students in the case the school does not participate in the program. Thus,

$$\begin{aligned} E[p_j^L] &= E[\tau_j 0 + (1 - \tau_j)p_j^0] \\ &= E[(1 - \tau_j)p_j^0] \end{aligned}$$

is the expected top-up fee schools charge to low income students under the targeted voucher program, which includes the mandated zero fees for participating schools ($\tau_j = 1$) and the sticker fee for non-participating schools ($\tau_j = 0$). Similarly,

$$E[p_j^H] = E[\tau_j p_j^1 + (1 - \tau_j)p_j^0]$$

is the expected top-up fee schools charge to higher income students under the targeted voucher program.

Using this notation, the effect of the reform on the average top-up fee schools charge to low income students is

$$E[p_j^L] - E[p_j] = E[(1 - \tau_j)p_j^0] - E[p_j]. \quad (2)$$

Likewise, the effect of the reform on the average top-up fee schools charge to higher income students is

$$E[p_j^H] - E[p_j] = E[\tau_j p_j^1 + (1 - \tau_j)p_j^0] - E[p_j]. \quad (3)$$

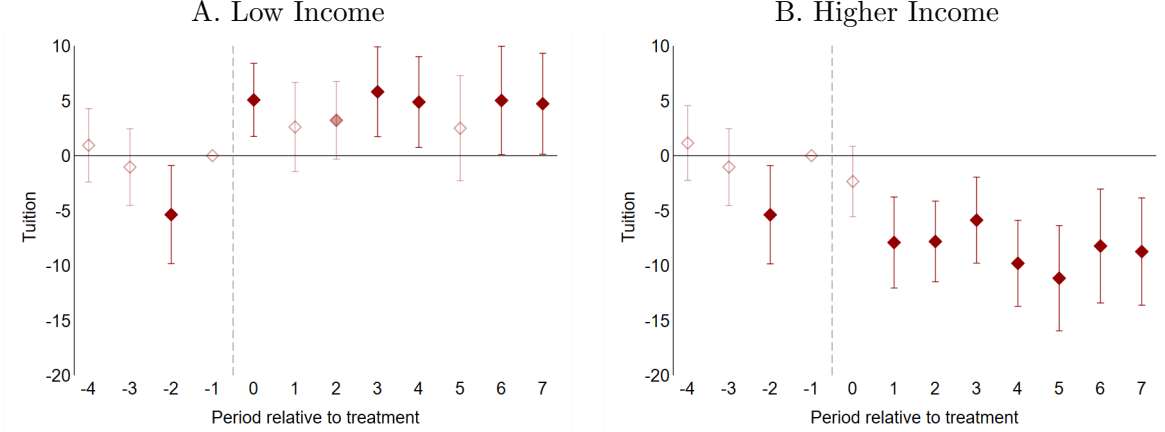
Figure 2 presents the effects of the targeted voucher reform on the average top-up fees schools charge to low income students (panel A) and on the average top-up fees schools charge to higher income students (panel B).¹¹ Pre-trends are in general parallel, except in period -2 . Panel A shows that on average private-voucher schools respond to the reform by increasing the top-up fee they charge to low income students. Such response is immediate, statistically significant, and stays somewhat constant over the period of study, at about \$3–6. Given that participating schools are mandated to charge no fee to low income students, this increase in the average fee result is necessarily driven by the counterfactual fees schools charge in the case they do not participate in the program, as is shown in equation (2).

¹⁰Recall that participating schools are mandated to charge \$0 to low income students.

¹¹Table B.1 in Appendix B presents the corresponding point estimates and standard errors.

Panel B shows that the reform induces private-voucher schools to lower the top-up fees they charge to higher income students, on average. The estimates show a quick and fairly constant reaction, of about \$8–11 reduction in fees. This response represents 45–65% the average fee private-voucher schools charge in 2007.

Figure 2: Effects of the Targeted Voucher Program on Private-voucher Schools - Top-up Fees



Notes: Fee values are real, and are converted from Chilean pesos to US dollars according to the exchange rate of Ch\$686.52 per US dollar, as of May 16, 2016. Results come from the estimation of event study regressions that use school level data for the period 2004–2015. Point estimates and 95% confidence intervals are presented. Hollow diamonds denote statistical insignificance. Light colored diamonds denote statistical significance at the 10% level. Dark colored diamonds denote statistical significance at the 5% level.

To have a better understanding of our findings, we analyze them through the lens of equations (2) and (3). Our results imply that equation (2) is greater than zero, and equation (3) is less than zero, simultaneously. Equivalently,

$$E[(1 - \tau_j)(p_j^0 - p_j)] > E[\tau_j p_j] \geq 0 \quad (4)$$

$$E[\tau_j(p_j^1 - p_j)] < -E[(1 - \tau_j)(p_j^0 - p_j)] \leq 0, \quad (5)$$

which implies market segmentation in prices; that is, participating schools reduce the fees they charge to higher income students, while non-participating schools increase their top-up fees.

4.4 Discussion

The market segmentation result we document is in line with Sánchez (2022), who estimates an empirical model of school choice and supply to study sorting under the targeted voucher policy (and counterfactual alternatives) in Chile. The estimated model implies that the policy induces market segmentation: lower-quality schools disproportionately opt into SEP and reduce fees, while higher-quality schools are more likely to opt out and raise fees to screen families that are averse to participation. A key component of the model is a “stigma” effect attached to participating in the program, which is substantially stronger for higher-income students. According to the estimated preferences, high-income families with highly educated mothers are

willing to pay over \$18,000 to avoid enrolling in a participating school, whereas low-income families with mothers without a high school diploma are nearly indifferent between participating and non-participating schools, all else equal.

Theoretical models of school competition, such as Rothschild and White (1995), Epple and Romano (1998), and Epple et al. (2006), predict stratified education markets and rationalize segmentation through peer effects.¹² These models assume that learning externalities make students value the ability of their schoolmates, which in equilibrium generates sorting by ability across schools.

In the context of the targeted voucher reform in Chile, the peer effects mechanism would imply that participating schools, which are more likely to be attended by low-income students, would reduce the fees they charge to higher-income students as an effort to counterbalance the disutility experienced by higher-income students with the increase of low-income students in the school after the reform. Non-participating schools, which are more likely to be attended by higher-income students, would increase the fees they charge to all students because they are relatively more attractive to these students after the reform. This strategic complementarity between prices and peers has been previously documented in the education literature (Allende, 2019).

Taken together, the literature and our findings suggest that peer effects are likely to be an important driver of the market segmentation in prices that we document. At the same time, peer effects can be viewed as a microfoundation for the “stigma” effect in Sánchez (2022), because the disutility that higher-income students experience from attending participating schools is plausibly related to expected peer composition.

To have a first insight on the plausibility of the peer effects mechanism being a driver of the market segmentation result that we find, we use pre-reform data to estimate a conventional school choice model, where preferences for peers are present (Allende, 2019; Crema, 2024). In such a model, a necessary condition for the peer effects mechanism to be a determinant of market segmentation is to find a statistically significant estimated preference parameter denoting disutility from low-income peers.

We follow Allende (2019), Sánchez (2022), Crema (2024), and Neilson (2025) and estimate a school choice model with peer effects using data for 2007, one year prior to the introduction of the targeted voucher reform. In the model, families choose among schools based on school characteristics, including the proportion of low-income students and top-up fees. Appendix C describes the model, the estimation procedure, and the results in detail.

Table C.3 in Appendix C shows that the preference parameter on the share of low-income students is negative and statistically significant for all family groups—both low- and high-income—and across all levels of mother’s education. This result indicates that all groups derive a statistically significant disutility from low-income peers. The magnitudes of the estimated parameters imply that families would need to be compensated by about \$877–\$2,910 to attend a school where all students are socioeconomically disadvantaged, relative to a school with no low-income peers (Table 4). We also observe a clear gradient by family background: families with higher socioeconomic status (proxied by mother’s education) require a larger compensation to attend schools with more low-income peers. Remarkably, even low-income families with mothers without a high school diploma require a substantial compensation (about \$877) to attend a school with exclusively

¹²An alternative mechanism for endogenous stratification is proposed by MacLeod and Urquiola (2015), who model students as valuing school reputation because employers infer productivity from school quality. This mechanism is less directly applicable in our setting because this stage of human capital formation precedes college entry, which in Chile is determined by admission exam scores that primarily reflect accumulated learning rather than a pure reputation signal.

low-income peers.

Table 4: Willingness to Pay for Low-Income Peers

	low-income	high-income
mother’s education: less than high school	−877	−1,415
mother’s education: high school	−1,264	−2,371
mother’s education: more than high school	−1,387	−2,910
mother’s education: missing	−889	−1,761

Notes: Willingness to pay estimates are calculated as the ratio of the estimated preference parameter for the share of low-income students and the estimated preference parameter for the effective price of attending the school in the school choice model described in Appendix C. Fees are in real USD 1,000 for the year 2015, using the exchange rate as of March 1, 2015 (648.88 CLP/USD).

In sum, our results are in line with theoretical models of school competition and stratification in which strong preferences for peers are a key driver of market segmentation in education markets.

5 Conclusions

This paper studies how a large targeted voucher reform affects the equilibrium organization of local education markets. We focus on Chile’s *Subvención Escolar Preferencial (SEP)*, which increased per-student funding for economically disadvantaged students while restricting schools’ ability to charge top-up fees to eligible students. Using administrative data covering the universe of schools and students over 2004–2015, we document that the reform generated sizable changes in the distribution of prices faced by families and contributed to segmentation of private-voucher markets along socioeconomic lines.

Our empirical analysis shows that private-voucher schools responded to the targeted voucher reform with tuition-setting decisions that differed sharply between participants and non-participants. Participating schools charged zero fees to eligible low-income students, as mandated by the program, while simultaneously lowering the fees faced by non-eligible students. By contrast, schools that opted out of the program increased posted fees. In the aggregate, these responses imply that low-income households experienced an increase in the average fee faced in the market, while higher-income households experienced a decrease in the average fee. The resulting pattern is consistent with the idea that targeted subsidies can, through schools’ strategic responses, reshape the effective price schedule faced by different groups of families and lead to re-sorting across schools.

We interpret these findings through the lens of school competition in markets where households value peer composition. When higher-income families derive disutility from attending schools with a larger share of low-income peers, schools have incentives to adjust prices in ways that reinforce differentiation: schools serving (or expecting to serve) more disadvantaged students lower the tuition faced by higher-income students to offset peer-related disutility, while schools specializing in higher-income students can raise fees as their demand becomes relatively less price elastic. The school choice evidence in Appendix C supports a key prerequisite for this mechanism: we estimate a statistically significant disutility from low-income peers across family groups, with willingness-to-pay magnitudes that rise with socioeconomic status.

From a policy perspective, our results underscore that evaluating targeted voucher programs requires

accounting for equilibrium supply-side responses. Chile’s targeted voucher design expanded resources for disadvantaged students and altered schools’ incentives, but it also induced price and composition adjustments that can affect who gains access to which schools and at what cost. These forces are likely to matter for welfare and for the distributional consequences of vouchers, especially when take-up is voluntary and schools can choose whether to segment.

Several limitations point to directions for future work. First, while we provide evidence on tuition responses and estimate preference parameters consistent with peer effects, quantifying the welfare consequences of the reform (including winners and losers across income groups) requires a full equilibrium model that jointly accounts for schools’ pricing and participation decisions and families’ sorting. Second, understanding longer-run dynamics—entry and exit, investment in quality, and the evolution of schools’ reputation—is important in a setting where accountability and monitoring interact with targeted funding. Finally, similar targeted programs are being considered and implemented in other countries (Epple et al., 2017; Abdulkadiroglu et al., 2018); applying the framework developed here to other institutional contexts would help isolate which features of program design are most effective at increasing equity without amplifying market segmentation.

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A Data

We combine various administrative data sets for Chilean students and schools for the years 2004–2015. Specifically, we use:

- Registry of students, 2004–2015.
These data provide information on students’ gender, date of birth, age, municipality of residence, type and level of education, grade, class, grade repetition status, special education status, and various characteristics of the school of attendance, such as municipality, type of administration (public, private-voucher, private-non-voucher), single/double shift schedule, and urban status.
- Registry of students’ academic performance, 2004–2015.
These data provide information on students’ gender, date of birth, municipality of residence, type and level of education, grade, class, GPA, average class attendance, and various characteristics of the school of attendance, such as municipality, type of administration, and urban status.
- Registry of schools, 2004–2015.
These data provide information on schools’ municipality, type of administration, urban status, address, and type and level of education offered.
- Registry of schools’ summary of enrollment, 2004–2015.
These data provide information on schools’ municipality, type of administration, urban status, male enrollment by education type and level, female enrollment by education type and level, total enrollment by education type and level, total enrollment, number of single-grade classes by education type and level, total number of single-grade classes, number of multigrade classes by education type and level, and total number of multigrade classes.
- Registry of teachers, 2004–2015.
These data provide information on teachers’ gender, date of birth, education degree, subject specialization, institution attended, graduation year, and duration of the degree studied. They also provide information on the characteristics of all schools in which each teacher is hired (municipality, type of management, rural status), and on the teachers’ primary and secondary roles (e.g. teacher, principal, supervisor), type of contract, hours contracted, teaching hours, experience, tenure, and teaching subject and level of education.
- Registry of schools that participate in the targeted voucher program, 2008–2015.
These data provide information on the characteristics of schools that participate in the targeted voucher program. Information on schools’ municipality, type of administration, urban status, targeted voucher classification, number of low income students that are eligible to benefit from the targeted voucher subsidy, and number of targeted voucher beneficiary students is also available.
- Registry of students that are eligible to participate in the targeted voucher program, 2008–2015.
These data provide information on the characteristics of low income students that are eligible to be beneficiaries of the targeted voucher program. They provide information on students’ gender, date of birth, targeted voucher participation status, level of education, grade, single/double shift schedule, and on the type of administration, urban status, and targeted voucher category of the school attended by the student.

- Private-voucher schools' top-up fees, 2004–2015.

These databases list all private-voucher schools that charge strictly positive top-up fees and indicate the amount of such fees.

B Additional Tables

Table B.1: Effects of the Targeted Voucher Program on Private-voucher School Top-up Fees

	low income	higher income
	(1)	(2)
-4	0.941 (1.700)	1.148 (1.724)
-3	-1.040 (1.773)	-1.039 (1.776)
-2	-5.368 (2.269)	-5.384 (2.270)
0	5.071 (1.693)	-2.352 (1.630)
1	2.605 (2.056)	-7.909 (2.104)
2	3.216 (1.796)	-7.815 (1.866)
3	5.808 (2.077)	-5.871 (1.987)
4	4.876 (2.100)	-9.798 (1.989)
5	2.493 (2.429)	-11.158 (2.434)
6	5.020 (2.505)	-8.227 (2.634)
7	4.722 (2.336)	-8.734 (2.480)
observations	38,271	38,271
R^2	0.272	0.218

Notes: Fee values are real, and are converted from Chilean pesos to US dollars according to the exchange rate of Ch\$686.52 per US dollar, as of May 16, 2016. Results come from the estimation of event study regressions that use school data from four years prior to the introduction of the 2008 targeted voucher program, and from the following eight years. Column (1) shows the effects of the reform on the top-up fees faced by low income students that are eligible to participate in the program, and column (2) shows the effects of the reform on the top-up fees faced by higher income students. Clustered (at the municipality level) standard errors are in parentheses.

C A Structural Model of School Choice Prior to the Reform

C.1 The Model

We develop a structural model of demand for schools motivated by Chile’s primary education system prior to the introduction of the targeted voucher reform. The model is static. There are several geographically separated education markets. Each market is populated by households that live in different locations within the market and have one primary-school-aged child. Subject to its budget constraint, each household chooses among the schools available in the market.

Students have heterogeneous preferences over schools’ fees, quality (i.e. a measure of how much the school increases students’ test scores), the share of low-income enrollment, geographic proximity, and a set of schools’ observable and unobservable (to the econometrician) characteristics. We capture preference heterogeneity with coefficients that vary with students’ observed demographic characteristics. Formally, in each market $m \in \{1, \dots, M\}$, student $i \in \{1, \dots, I\}$ chooses the school $j \in \{1, \dots, J\}$ that maximizes her utility. We specify the student’s conditional indirect utility as

$$U_{ijt} = \beta_{1i}p_{jt} + \beta_{2i}d_{ijt} + \beta'_{3i}X_{it} + \beta_{4i}^{\zeta}q_{jt} + \beta_{5i}^{\zeta}\hat{z}_{jt}^{y^L} + \xi_{jt}^{\zeta} + \varepsilon_{ijt}$$

where the superscript $\zeta \in \{L, H\}$ refers to whether the student is low-income or high-income, and the subscript t indexes markets.¹³ Thus, p_{jt} is school j ’s top-up fee, d_{ijt} is the distance from student i ’s home to school j , X_{jt} is a vector of school j ’s observable characteristics, q_{jt} is school j ’s quality measure, $\hat{z}_{jt}^{y^L}$ is the expected share of low-income enrollment at school j , ξ_{jt}^{ζ} is the common preference that students of type ζ have for school j ’s unobservable (to the econometrician) characteristics, and ε_{ijt} is an i.i.d. preference shock. Also, for any $\beta^{\zeta} \in \{\beta_4^{\zeta}, \xi_{jt}^{\zeta}\}$, let $\beta^{\zeta} = D_i\beta^L + (1 - D_i)\beta^H$, where $D_i = \mathbb{1}[i \text{ is low-income}]$. Similarly, for $k = 1, 2, 3, 5$, $\beta_{kit} = D_i\beta_{kit}^L + (1 - D_i)\beta_{kit}^H$, where $\beta_{kit}^L = \beta_k^L + \sum_r z_{ir}\beta_{krt}^L$ and $\beta_{kit}^H = \beta_k^H + \sum_r z_{ir}\beta_{krt}^H$, with z_{ir} denoting student i ’s demographic characteristic r .

Let $V_{ijt} = \beta_{1i}p_{jt} + \beta_{2i}d_{ijt} + \beta'_{3i}X_{it} + \beta_{4i}^{\zeta}q_{jt} + \beta_{5i}^{\zeta}\hat{z}_{jt}^{y^L} + \xi_{jt}^{\zeta}$. Thus, $U_{ijt} = V_{ijt} + \varepsilon_{ijt}$. Assuming $\varepsilon_{ijt} \sim \text{Type I Extreme Value}$, the probability that student i chooses school j is logistic:

$$P_{ijt} = \frac{e^{V_{ijt}}}{\sum_k e^{V_{ikt}}}.$$

C.2 Estimation and Identification

We work with non-overlapping geographic markets, closely following Sánchez (2022) and Neilson (2025). For estimation, we proceed in three steps. First, we estimate school popularity fixed effects and the parameters of heterogeneity in students’ preferences. Then, given that school quality is not directly observed in the data, we obtain a school quality measure by estimating test scores equations using a selection-on-observables approach, very similar to the work in Allende (2019), Allende et al. (2019), and Neilson (2025). Finally, we link school quality and other characteristics to school popularity estimates to finalize preferences for schools estimation.

¹³The low/high-income classification follows the eligibility criteria for the targeted voucher program and is heavily determined by household income.

Preferences for Schools. We use Maximum Likelihood to estimate preferences for proximity, taste heterogeneity, and mean utilities or school popularity. Heterogeneity in preferences is captured by a set of students' observable demographic characteristics, which in practice are mother's educational level dummies. Mean utilities vary at the student's socioeconomic status (low/high-income), and absorb the remaining preference components from the indirect utility function,

$$\delta_{jt}^\zeta = \beta_1^\zeta p_{jt} + \beta_3^{\zeta'} X_{jt}^\beta + \beta_4^\zeta q_{jt} + \beta_5^\zeta \hat{z}_{jt}^{y^L} + \xi_{jt}^\zeta.$$

The corresponding log-likelihood function is:

$$LL(\beta) = \sum_i \sum_j e_{ij} \ln \left(\frac{\exp \left((\beta_{1i}^\zeta - \beta_1^\zeta) p_{jt} + \beta_{2i} d_{ijt} + (\beta_{3i}^\zeta - \beta_3^\zeta) X_{jt}^\beta + (\beta_{5i}^\zeta - \beta_5^\zeta) \hat{z}_{jt}^{y^L} + \delta_{jt}^\zeta \right)}{\sum_k \exp \left((\beta_{1i}^\zeta - \beta_1^\zeta) p_{kt} + \beta_{2i} d_{ikt} + (\beta_{3i}^\zeta - \beta_3^\zeta) X_{kt}^\beta + (\beta_{5i}^\zeta - \beta_5^\zeta) \hat{z}_{kt}^{y^L} + \delta_{kt}^\zeta \right)} \right),$$

where e_{ij} is the choice indicator, i.e. $e_{ij} = 1$ indicates that student i attends school j .

School Quality. To estimate the measure of school quality, we follow Allende (2019), Allende et al. (2019), and Neilson (2025), and estimate the parameters that enter an auxiliary test scores equation using an offline selection-on-observables linear model. Note that school value-added measures, as defined by the value-added literature (Rockoff, 2004; Rivkin et al., 2005; Kane and Staiger, 2008; Chetty et al., 2014a,b; Bau, 2022), are not possible to estimate in this context, because prior test scores are not observed. Instead, our school quality estimates measure the accumulated quality value of a school for grades 1st–4th.

In practice, we estimate by OLS the following regression:

$$Y_{ij} = W_i \gamma + \sum_j q_j e_{ij} + \epsilon_i,$$

where we include a rich set of observables in W_i to capture as much variation as possible and mitigate inconsistency from selection on observables. The covariates include gender, low-income status, grade repetition, mother's education, father's education, household income, having attended pre-K, having attended kindergarten, computer availability at home, internet availability at home, indigenous status, number of books at home, and class attendance. We use the student-level average of 4th grade national standardized scores in verbal and math as the outcome variable. This set of observables is in line with Allende (2019), Allende et al. (2019), and Neilson (2025). Moreover, Allende et al. (2019) show that school quality estimates based on this procedure and a similar set of covariates yield results comparable to those obtained when adding prior test scores as controls; they also provide additional experimental evidence supporting the strategy. We estimate the test score equations market by market and obtain $\{\hat{q}_j\}_{j=1}^J$ as school quality estimates.

Linking Mean Utilities to School Quality and Other Characteristics. We use the estimated $\hat{\delta}_j^\zeta$ terms from the first step, as well as the $\hat{q}_j^*$ shrunk estimated school quality from the test scores regressions to estimate the remaining mean preference parameters in a linear regression of the form:

$$\hat{\delta}_{jt}^\zeta = \beta_1^\zeta p_{jt} + \beta_3^{\zeta'} X_{jt}^\beta + \beta_4^\zeta \hat{q}_{jt}^* + \beta_5^\zeta \hat{z}_{jt}^{y^L} + \xi_{jt}^\zeta. \quad (6)$$

As is usual in demand models, we assume that X_{jt}^β is uncorrelated with ξ_{jt}^ζ . However, we allow p_{jt} and $\hat{z}_{jt}^{y^L}$ to

be endogenous—i.e. correlated with ξ_j^ζ —and estimate equation (6) by 2SLS using the instruments described below. To account for the estimation noise in the dependent variable, $\hat{\delta}_j^\zeta$, we weigh each observation by the inverse of the maximum likelihood-estimated variance of the school fixed effect.

Identification and Instruments. The term ξ_j^ζ represents type- ζ families’ preferences for school j ’s unobserved characteristics, such as principal’s charisma or managerial style, educational curriculum, etc., that may be correlated with school’s choices for top-up fees and with its student body composition. An artificial example is a school with a mission to serve vulnerable families. Such school may be willing to develop a culture and an educational environment that are particularly welcoming to low-income students, which may intensify the school’s propensity to enroll vulnerable students and to set low fees. Thus, consistent estimation of the linear parameters in equation (6) necessitates instruments.

Our choice of instruments for program participation and top-up fees follows related studies (Allende, 2019; Allende et al., 2019; Sánchez, 2022; Neilson, 2025). For prices, we use the average per student universal voucher received by the school, other per student non-voucher subsidies (e.g. teacher incentives, rurality bonus), and the local share of private competitors. For schools’ student body composition, we use the local share of students with a highly educated mother. For the local shares of private competitors and students with a highly educated mother, we construct distance-weighted versions of these variables, where the weights are the inverse of the distance to the school. This way, we capture the local market structure and demographic composition that are more relevant for each school.

Note that instruments related to market structure and demographic characteristics are relevant in a context in which students do not travel long distances to attend school. Taking residential sorting and market structure as given, a school located in a neighborhood with a high concentration of high-income families and low competition from other private schools should enroll a higher share of high-income students than a school located in a neighborhood with a lower concentration of high-income families and more private-school competition.

C.3 Results

C.3.1 School Quality

We estimate school quality under a selection-on-observables assumption (Allende, 2019; Allende et al., 2019; Sánchez, 2022; Neilson, 2025), running test score regressions on a large set of observable characteristics and school indicators, market by market. We use the student-level average of 4th grade national standardized scores in verbal and math. The covariates include gender, low-income status, grade repetition, mother’s education, father’s education, household income, having attended pre-K, having attended kindergarten, computer availability at home, internet availability at home, indigenous status, and number of books at home. For brevity, we do not present each market’s estimates, but they are available upon request.

C.3.2 Preferences

First Stage. As stated in Section C.2, we estimate the first-stage regressions for the endogenous variables (top-up fees and the share of low-income students in the school) using instrumental variables and school characteristics.

Table C.2 presents the first-stage results. Local competition (measured by the pre-reform local share of private competitors) is associated with lower fees (-0.041) and a higher share of low-income students (0.010). Mother’s education (more than high school) is positively associated with top-up fees (0.265) and negatively associated with the share of low-income students (-0.158), consistent with sorting patterns in which more educated families select into higher-fee schools with fewer low-income peers. The average per-student universal voucher is negatively associated with fees (-0.027) but positively associated with the share of low-income students (0.118), suggesting that schools receiving larger universal voucher subsidies attract more low-income students while keeping fees relatively low. Non-voucher subsidies have virtually no association with fees (coefficient near zero) but a positive association with student composition (0.039), suggesting that schools receiving more subsidies attract more low-income students. Public schools charge significantly lower fees (-0.203) and serve more low-income students (0.073), consistent with their mission. Rural schools charge lower fees (-0.058) and serve more low-income populations (0.116). The quality indicator is positively associated with fees (0.069) and negatively associated with the low-income student share (-0.032), consistent with a quality–price gradient and cream-skimming. The F-statistics for excluded instruments (71.44 for fees and 87.80 for share) are well above standard thresholds, confirming instrument relevance.

Table C.2: First Stage Estimates for 2007

	top-up fees		share of low income students	
	coef.	std. err.	coef.	std. err.
pre-reform local share of private competitors	-0.041	0.014	0.010	0.010
Mother’s education: more than high school	0.265	0.016	-0.158	0.012
avg. per student universal voucher	-0.027	0.038	0.118	0.028
avg. per student non-voucher subsidies	-0.000	0.005	0.039	0.003
full day shift	0.066	0.011	-0.019	0.008
secular	0.001	0.007	-0.022	0.005
rural	-0.058	0.009	0.116	0.006
public	-0.203	0.009	0.073	0.007
quality	0.069	0.006	-0.032	0.004
intercept	0.193	0.026	0.108	0.019
F excluded IV	71.44		87.80	
no. of schools			3,080	

Notes: Results from the first stage of the 2SLS estimation of mean preference parameters. Mother’s education and the local share of private schools are distance weighted. We deflated 2007 fee amounts to 2015 values. Fee amounts are in real \$1,000 (2015) and were converted from CLP to USD using the exchange rate as of March 1, 2015 (648.88 CLP/USD). * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Preference parameters. We present estimates of families’ preferences in Table C.3, separately for low- and high-income groups. Within each income group, preferences for most school characteristics vary by mother’s education, where the omitted category is “more than high school.” Low-income families are substantially more price sensitive than higher-income families (coefficients of -11.195 and -6.668 , respectively). There is also meaningful heterogeneity within each income group: families with less educated mothers are more responsive to price changes than families with more educated mothers. Both low- and high-income

families exhibit a strong negative preference for schools with higher shares of low-income students (coefficients of -15.524 and -19.406 , respectively), with important variation by mother’s education. Interestingly, low-income families display a negative valuation of school quality (coefficient -0.381), whereas high-income families value school quality positively (coefficient 0.610), suggesting a divergence in how quality is perceived across income groups during this period. Full-day-shift schools are positively valued by low-income families (coefficient 1.539) but less so by high-income families. Secular and public schools are generally disfavored by both income groups. Both groups prefer schools that are closer to their homes, with stronger distance sensitivity among low-income families (distance coefficients of -0.114 and -0.071 , respectively). Overall, the estimates highlight substantial heterogeneity in school choice preferences across socioeconomic groups and mother’s education.

Table C.3: Estimates from Demand Model for 2007

	Low-Income		High-Income	
	Coef.	Std. Err.	Coef.	Std. Err.
Fees	-11.195	1.383	-6.668	1.046
Fees $\times$ Mother's education: less than high school	-4.713	0.259	-3.443	0.079
Fees $\times$ Mother's education: high school	-2.104	0.223	-1.226	0.047
Fees $\times$ Mother's education: missing	-3.573	0.381	-0.990	0.092
Share low income students	-15.524	1.707	-19.406	1.292
Share low income students $\times$ Mother's education: less than high school	1.568	0.262	5.102	0.185
Share low income students $\times$ Mother's education: high school	-1.284	0.268	0.688	0.167
Share low income students $\times$ Mother's education: missing	2.391	0.299	5.920	0.242
Quality	-0.381	0.115	0.610	0.087
Full day shift	1.539	0.174	0.351	0.132
Full day shift $\times$ Mother's education: less than high school	-0.541	0.071	0.035	0.035
Full day shift $\times$ Mother's education: high school	-0.515	0.072	0.117	0.029
Full day shift $\times$ Mother's education: missing	-0.621	0.086	-0.164	0.051
Secular	-0.316	0.110	-0.166	0.084
Secular $\times$ Mother's education: less than high school	-0.137	0.054	-0.033	0.028
Secular $\times$ Mother's education: high school	-0.096	0.055	-0.056	0.023
Secular $\times$ Mother's education: missing	-0.190	0.063	0.061	0.039
Rural	-0.023	0.228	-0.535	0.172
Rural $\times$ Mother's education: less than high school	0.313	0.079	0.131	0.054
Rural $\times$ Mother's education: high school	0.082	0.082	-0.172	0.051
Rural $\times$ Mother's education: missing	0.264	0.089	0.049	0.069
Public	-0.727	0.237	-0.354	0.179
Public $\times$ Mother's education: less than high school	-0.009	0.064	-0.025	0.037
Public $\times$ Mother's education: high school	-0.044	0.065	0.007	0.033
Public $\times$ Mother's education: missing	-0.006	0.075	-0.017	0.050
Distance (interact)	-0.114	0.002	-0.071	0.002
Distance (non-missing)	0.360	0.067	0.263	0.041
No. of schools	3,038			
No. of students	100,714			

Notes: Results from maximum likelihood estimation of distance and preference heterogeneity by mother's education, and from 2SLS estimation of mean preference parameters. Omitted mother's level of education category is "more than high school". The distance from home to school variables are the interaction of a dummy for non-missing and the corresponding distance variables. The non-missing dummy variable is also added in the model. Distance is measured in 100 meters. We deflated 2007 fee amounts to 2015 values. Fee amounts are in real \$1,000 for the year 2015, and were transformed from CLP to USD according to the exchange rate as of March 1, 2015 (648.88 CLP/USD).

Willingness to pay for school attributes. To complement the demand estimates, we compute families' willingness to pay (WTP) for school attributes by dividing the relevant coefficients by the negative of the price coefficient, for each income group and mother's education level (Berry et al., 1995; Petrin,

2002). Table C.4 presents these results. Both low- and high-income families exhibit negative WTP for higher shares of low-income students across all education categories. For instance, high-income families with mothers with at least a high school diploma require compensation of \$2,910 to attend a school with exclusively disadvantaged students, while low-income families with the same education profile require \$1,387. The quality dimension follows the estimation results: low-income families require compensation to attend a higher-quality school, which may reflect identification challenges. In contrast, high-income families show positive but modest WTP for quality (ranging from \$60 to \$91). For secular schools, both income groups require compensation across all mother's education categories (ranging from \$14 to \$34), consistent with preferences for religious schools. To attend a public school, families require compensation across all heterogeneity groups (ranging from \$38 to \$65). Full-day-shift schools have positive WTP, with low-income families particularly valuing this attribute (WTP ranging from \$63 to \$137). Rural schools show mixed patterns: low-income families exhibit small positive or near-zero WTP, while high-income families require compensation (ranging from \$40 to \$90), suggesting stronger urban preferences among higher-SES households. Finally, all families are willing to pay to attend nearer schools, all else equal.

Table C.4: Willingness to Pay for School Attributes - 2007

	low income	high income
<i>Share of low income students</i>		
mother's education: less than high school	-877	-1,415
mother's education: high school	-1,264	-2,371
mother's education: more than high school	-1,387	-2,910
mother's education: missing	-889	-1,761
<i>Quality</i>		
mother's education: less than high school	-24	60
mother's education: high school	-29	77
mother's education: more than high school	-34	91
mother's education: missing	-26	80
<i>Secular</i>		
mother's education: less than high school	-28	-20
mother's education: high school	-31	-28
mother's education: more than high school	-28	-25
mother's education: missing	-34	-14
<i>Public</i>		
mother's education: less than high school	-46	-38
mother's education: high school	-58	-44
mother's education: more than high school	-65	-53
mother's education: missing	-50	-49
<i>Full day shift</i>		
mother's education: less than high school	63	38
mother's education: high school	77	59
mother's education: more than high school	137	53
mother's education: missing	62	24
<i>Rural</i>		
mother's education: less than high school	18	-40
mother's education: high school	4	-90
mother's education: more than high school	-2	-80
mother's education: missing	16	-64
<i>Distance</i>		
mother's education: less than high school	-7	-7
mother's education: high school	-9	-9
mother's education: more than high school	-10	-11
mother's education: missing	-8	-9

Notes: Willingness-to-pay (WTP) estimates are calculated by dividing the corresponding estimated coefficients by the negative of the price coefficient, for each income group and mother's education level. 2007 fee amounts are deflated to 2015 values. Values are expressed in real 2015 \$1,000 (USD), using the exchange rate as of March 1, 2015 (648.88 CLP/USD). Distance is measured in 100 meters.